

# Probability and Statistics

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## Lecture 28 **Hypothesis Testing**

**Muhammad Mahbub Alam**

Professor  
CSE Department, IUT

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# Hypothesis Testing

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A statistical hypothesis is usually a statement about a set of parameters of a population distribution. It is called a hypothesis because it is not known whether or not it is true. A primary problem is to develop a procedure for de-

Note that in accepting a given hypothesis we are not actually claiming that it is true but rather we are saying that the resulting data appear to be consistent with it. For instance, in the case of a normal  $(\theta, 1)$  population, if a resulting sample of size 10 has an average value of 1.25, then although such a result cannot be regarded as being evidence in favor of the hypothesis " $\theta < 1$ ," it is not inconsistent with this hypothesis, which would thus be accepted. On the

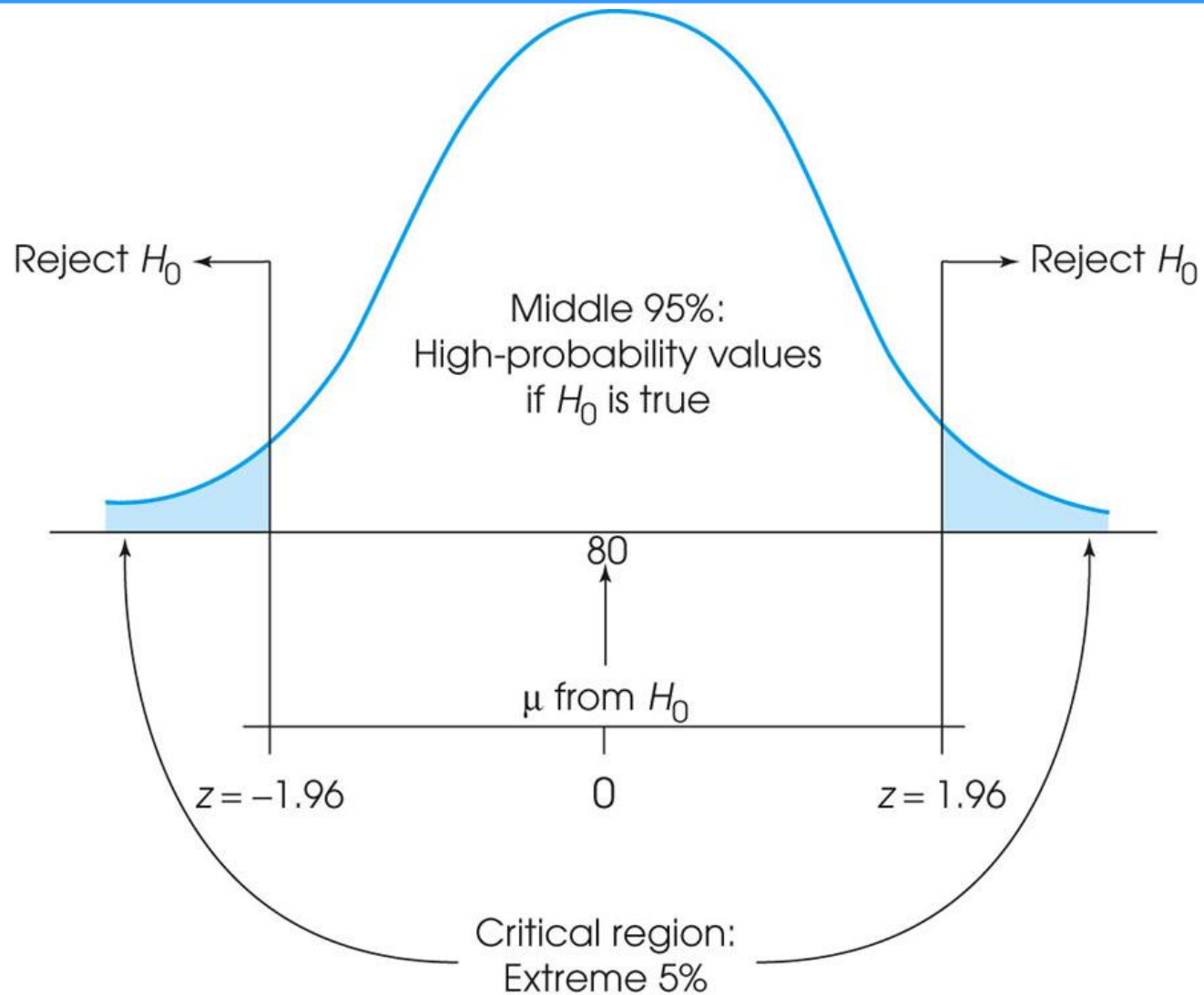
Consider a population having distribution  $F_\theta$ , where  $\theta$  is unknown, and suppose we want to test a specific hypothesis about  $\theta$ . We shall denote this hypothesis by  $H_0$  and call it the *null hypothesis*. For example, if  $F_\theta$  is a normal distribution function with mean  $\theta$  and variance equal to 1, then two possible null hypotheses about  $\theta$  are

(a)  $H_0 : \theta = 1$

(b)  $H_0 : \theta \leq 1$

Thus the first of these hypotheses states that the population is normal with mean 1 and variance 1, whereas the second states that it is normal with variance 1 and a mean less than or equal to 1. Note that the null hypothesis in (a),

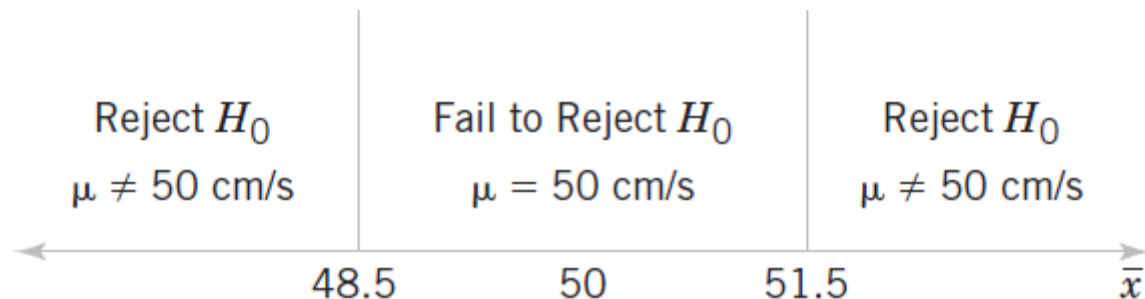
hypothesis in (b) does not. A hypothesis that, when true, completely specifies the population distribution is called a *simple* hypothesis; one that does not is called a *composite* hypothesis.



# Hypothesis Testing

$H_0: \mu = 50$  centimeters per second

$H_1: \mu \neq 50$  centimeters per second



## Type I Error

Rejecting the null hypothesis  $H_0$  when it is true is defined as a **type I error**.

$$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$$

## Type II Error

Failing to reject the null hypothesis when it is false is defined as a **type II error**.

$$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$$

# Decisions in Hypothesis Testing

| Decision             | $H_0$ Is True | $H_0$ Is False |
|----------------------|---------------|----------------|
| Fail to reject $H_0$ | No error      | Type II error  |
| Reject $H_0$         | Type I error  | No error       |

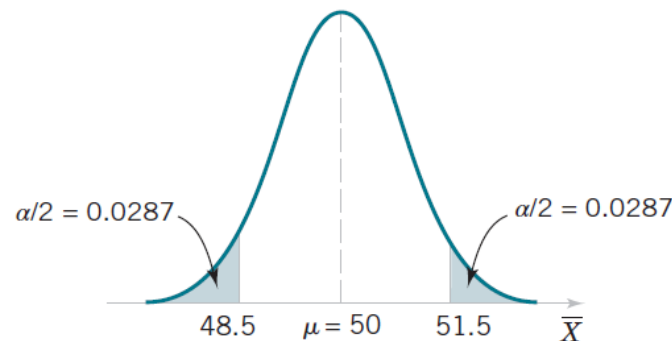
$$\alpha = P(\bar{X} < 48.5 \text{ when } \mu = 50) + P(\bar{X} > 51.5 \text{ when } \mu = 50)$$

The  $z$ -values that correspond to the critical values 48.5 and 51.5 are

$$z_1 = \frac{48.5 - 50}{0.79} = -1.90 \quad \text{and} \quad z_2 = \frac{51.5 - 50}{0.79} = 1.90$$

Therefore,

$$\alpha = P(z < -1.90) + P(z > 1.90) = 0.0287 + 0.0287 = 0.0574$$



## 8.3 Tests concerning the mean of a normal population

### 8.3.1 Case of known variance

$$H_0 : \mu = \mu_0$$

against the alternative hypothesis

$$H_1 : \mu \neq \mu_0$$

where  $\mu_0$  is some specified constant.

Since  $\bar{X} = \sum_{i=1}^n X_i / n$  is a natural point estimator of  $\mu$ , it seems reasonable to accept  $H_0$  if  $\bar{X}$  is not too far from  $\mu_0$ . That is, the critical region of the test would be of the form

$$C = \{X_1, \dots, X_n : |\bar{X} - \mu_0| > c\} \tag{8.3.1}$$

for some suitably chosen value  $c$ .



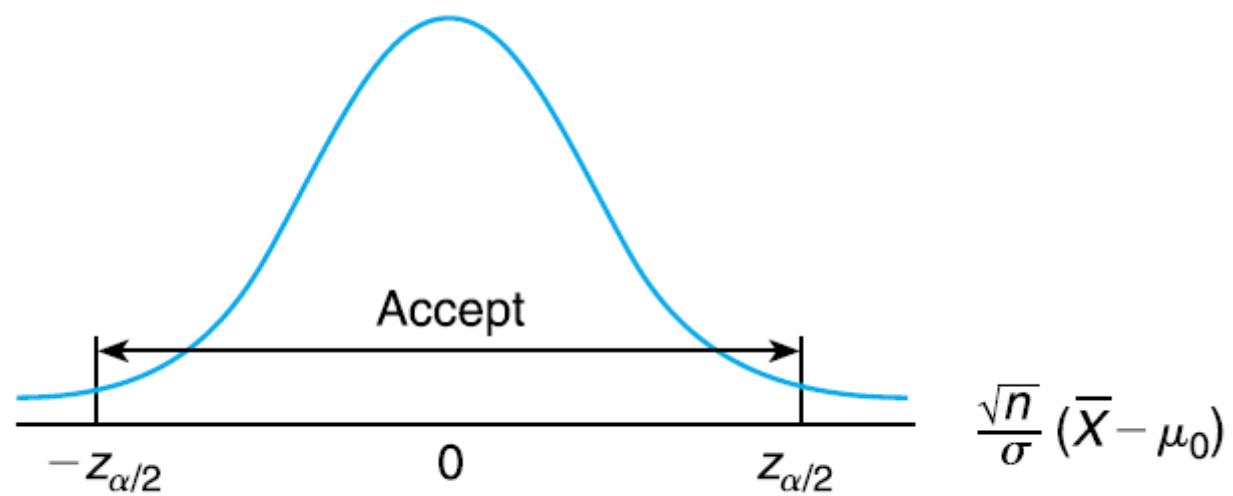
$$P_{\mu_0}\{|\bar{X} - \mu_0| > c\} = \alpha \quad (8.3.2)$$

where we write  $P_{\mu_0}$  to mean that the preceding probability is to be computed under the assumption that  $\mu = \mu_0$ . However, when  $\mu = \mu_0$ ,  $\bar{X}$  will be normally distributed with mean  $\mu_0$  and variance  $\sigma^2/n$  and so  $Z$ , defined by

$$Z \equiv \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

Thus, the significance level  $\alpha$  test is to reject  $H_0$  if  $|\bar{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$  and accept otherwise; or, equivalently, to

$$\begin{array}{lll} \text{reject} & H_0 & \text{if } \frac{\sqrt{n}}{\sigma}|\bar{X} - \mu_0| > z_{\alpha/2} \\ \text{accept} & H_0 & \text{if } \frac{\sqrt{n}}{\sigma}|\bar{X} - \mu_0| \leq z_{\alpha/2} \end{array} \quad (8.3.3)$$



### 9.1.6

## General Procedure for Hypothesis Tests

This chapter develops hypothesis-testing procedures for many practical problems. Use of the following sequence of steps in applying hypothesis-testing methodology is recommended.

1. **Parameter of interest:** From the problem context, identify the parameter of interest.
2. **Null hypothesis,  $H_0$ :** State the null hypothesis,  $H_0$ .
3. **Alternative hypothesis,  $H_1$ :** Specify an appropriate alternative hypothesis,  $H_1$ .
4. **Test statistic:** Determine an appropriate test statistic.
5. **Reject  $H_0$  if:** State the rejection criteria for the null hypothesis.
6. **Computations:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
7. **Draw conclusions:** Decide whether or not  $H_0$  should be rejected and report that in the problem context.

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**Example 8.3.a.** It is known that if a signal of value  $\mu$  is sent from location A, then the value received at location B is normally distributed with mean  $\mu$  and standard deviation 2. That is, the random noise added to the signal is an  $N(0, 4)$  random variable. There is reason for the people at location B to suspect that the signal value  $\mu = 8$  will be sent today. Test this hypothesis if the same signal value is independently sent five times and the average value received at location B is  $\bar{X} = 9.5$ .

**Solution.** Suppose we are testing at the 5 percent level of significance. To begin, we compute the test statistic

$$\frac{\sqrt{n}}{\sigma} |\bar{X} - \mu_0| = \frac{\sqrt{5}}{2} (1.5) = 1.68$$

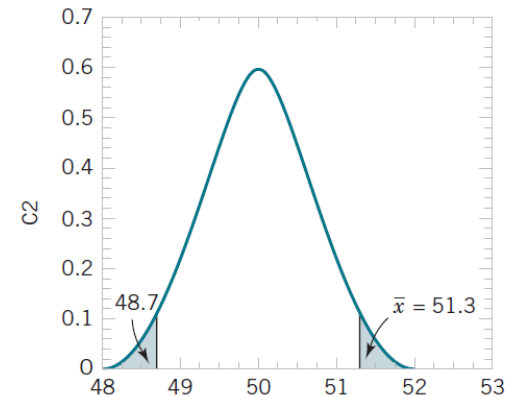
Since this value is less than  $z_{.025} = 1.96$ , the hypothesis is accepted. In other words, the data are not inconsistent with the null hypothesis in the sense that a sample average as far from the value 8 as observed would be expected, when the true mean is 8, over 5 percent of the time. Note, however, that if a less stringent significance level were chosen — say  $\alpha = .1$  — then the null hypothesis would have been rejected. This follows since  $z_{.05} = 1.645$ , which is less than 1.68. Hence, if we would have chosen a test that had a 10 percent chance of rejecting  $H_0$  when  $H_0$  was true, then the null hypothesis would have been rejected.

## P-Value

The **P-value** is the smallest level of significance that would lead to rejection of the null hypothesis  $H_0$  with the given data.

$$H_0: \mu = 50 \quad H_1: \mu \neq 50$$

$$\begin{aligned} P\text{-value} &= 1 - P(48.7 < \bar{X} < 51.3) \\ &= 1 - P\left(\frac{48.7 - 50}{2.5/\sqrt{16}} < Z < \frac{51.3 - 50}{2.5/\sqrt{16}}\right) \\ &= 1 - P(-2.08 < Z < 2.08) \\ &= 1 - 0.962 = 0.038 \end{aligned}$$



$$\begin{aligned} P(-2.08 < Z < 2.08) &= \text{chi}(2.08) - (1 - \text{chi}(2.08)) \\ &= \text{chi}(2.08) - (1 - 0.9812) \\ &= 0.9812 - 0.0188 = 0.9624. \end{aligned}$$

**Example 8.3.b.** In Example 8.3.a, suppose that the average of the 5 values received is  $\bar{X} = 8.5$ . In this case,

$$\frac{\sqrt{n}}{\sigma} |\bar{X} - \mu_0| = \frac{\sqrt{5}}{4} = .559$$

Since

$$\begin{aligned} P\{|Z| > .559\} &= 2P\{Z > .559\} \\ &= 2 \times .288 = .576 \end{aligned}$$

it follows that the  $p$ -value is .576 and thus the null hypothesis  $H_0$  that the signal sent has value 8 would be accepted at any significance level  $\alpha < .576$ . Since we would clearly never want to test a null hypothesis using a significance level as large as .576,  $H_0$  would be accepted.

On the other hand, if the average of the data values were 11.5, then the  $p$ -value of the test that the mean is equal to 8 would be

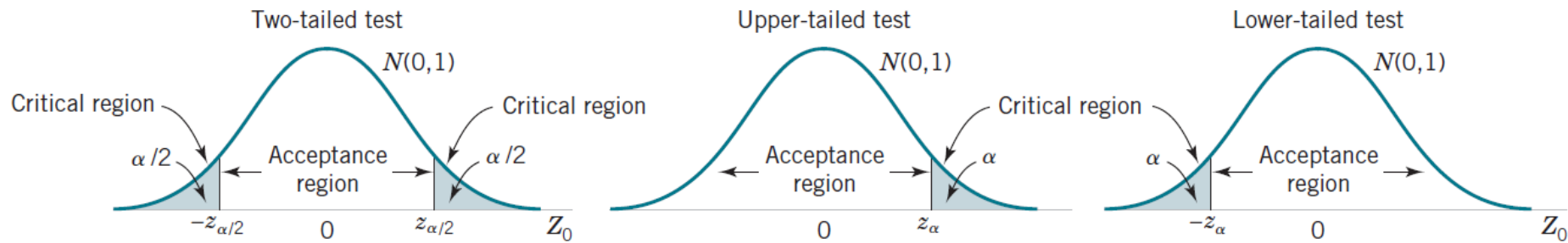
$$\begin{aligned} P\{|Z| > 1.75\sqrt{5}\} &= P\{|Z| > 3.913\} \\ &\approx .00005 \end{aligned}$$

For such a small  $p$ -value, the hypothesis that the value 8 was sent is rejected.

**Table 8.1**  $X_1, \dots, X_n$  Is a Sample from a  $(\mu, \sigma^2)$  Population,  $\sigma^2$  Is Known,  
 $\bar{X} = \sum_{i=1}^n X_i/n$ .

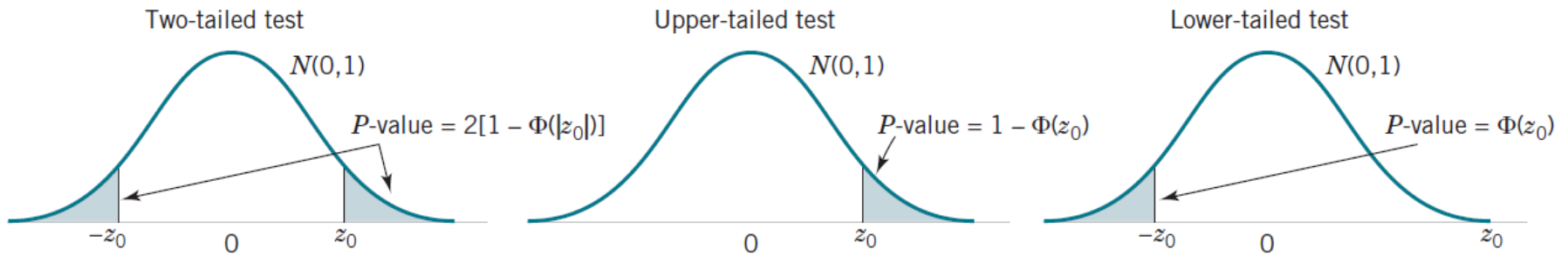
| $H_0$                                     | $H_1$            | Test Statistic $TS$                | Significance Level $\alpha$ Test | $p$ -Value if $TS = t$ |
|-------------------------------------------|------------------|------------------------------------|----------------------------------|------------------------|
| $\mu = \mu_0$                             | $\mu \neq \mu_0$ | $\sqrt{n}(\bar{X} - \mu_0)/\sigma$ | Reject if $ TS  > z_{\alpha/2}$  | $2P\{Z \geq  t \}$     |
| $\mu \leq \mu_0$                          | $\mu > \mu_0$    | $\sqrt{n}(\bar{X} - \mu_0)/\sigma$ | Reject if $TS > z_{\alpha}$      | $P\{Z \geq t\}$        |
| $\mu \geq \mu_0$                          | $\mu < \mu_0$    | $\sqrt{n}(\bar{X} - \mu_0)/\sigma$ | Reject if $TS < -z_{\alpha}$     | $P\{Z \leq t\}$        |
| $Z$ is a standard normal random variable. |                  |                                    |                                  |                        |



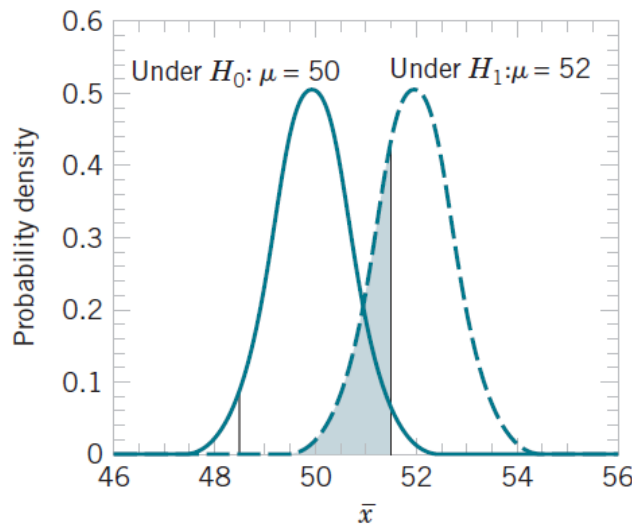


$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$P = 2 [1 - \Phi(|z_0|)]$$



# Type II Error



$H_0: \mu = 50$  centimeters

$H_1: \mu \neq 50$  centimeters

$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$

$\beta = P(48.5 \leq X \leq 51.5 \text{ when } \mu = 52)$

The  $z$ -values corresponding to 48.5 and 51.5 when  $\mu = 52$  are

$$z_1 = \frac{48.5 - 52}{0.79} = -4.43 \quad \text{and} \quad z_2 = \frac{51.5 - 52}{0.79} = -0.63$$

Therefore,

$$\begin{aligned} \beta &= P(-4.43 \leq Z \leq -0.63) = P(Z \leq -0.63) - P(Z \leq -4.43) \\ &= 0.2643 - 0.0000 = 0.2643 \end{aligned}$$

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$$\beta(\mu) = P_{\mu}\{\text{acceptance of } H_0\}$$

$$= P_{\mu} \left\{ \left| \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \right| \leq z_{\alpha/2} \right\}$$

$$= P_{\mu} \left\{ -z_{\alpha/2} \leq \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2} \right\}$$

$$\beta(\mu) = P_{\mu} \left\{ -z_{\alpha/2} \leq \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2} \right\}$$

$$= P_{\mu} \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq \frac{\bar{X} - \mu_0 - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\}$$

$$= P_{\mu} \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq Z - \frac{\mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\}$$

$$= P \left\{ \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq Z \leq \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right\}$$

$$= \Phi \left( \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right) - \Phi \left( \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \right)$$

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**Example 8.3.c.** For the problem presented in Example 8.3.a, let us determine the probability of accepting the null hypothesis that  $\mu = 8$  when the actual value sent is 10. To do so, we compute

$$\frac{\sqrt{n}}{\sigma}(\mu_0 - \mu) = -\frac{\sqrt{5}}{2} \times 2 = -\sqrt{5}$$

As  $z_{.025} = 1.96$ , the desired probability is, from Equation (8.3.4),

$$\begin{aligned}\Phi(-\sqrt{5} + 1.96) - \Phi(-\sqrt{5} - 1.96) \\&= 1 - \Phi(\sqrt{5} - 1.96) - [1 - \Phi(\sqrt{5} + 1.96)] \\&= \Phi(4.196) - \Phi(.276) \\&= .392 \quad \blacksquare\end{aligned}$$

The type II error probability also depends on the sample size  $n$ .

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 52)$$

When  $n = 16$ , the standard deviation of  $\bar{X}$  is  $\sigma/\sqrt{n} = 2.5/\sqrt{16} = 0.625$ , and the  $z$ -values corresponding to 48.5 and 51.5 when  $\mu = 52$  are

$$z_1 = \frac{48.5 - 52}{0.625} = -5.60 \quad \text{and} \quad z_2 = \frac{51.5 - 52}{0.625} = -0.80$$

Therefore,

$$\begin{aligned}\beta &= P(-5.60 \leq Z \leq -0.80) = P(Z \leq -0.80) - P(Z \leq -5.60) \\ &= 0.2119 - 0.0000 = 0.2119\end{aligned}$$

Recall that when  $n = 10$  and  $\mu = 52$ , we found that  $\beta = 0.2643$

increasing the sample size results in a decrease in the probability of type II error.

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$$n \approx \frac{(z_{\alpha/2} + z_{\beta})^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

**Example 8.3.d.** For the problem of Example 8.3.a, how many signals need be sent so that the .05 level test of  $H_0 : \mu = 8$  has at least a 75 percent probability of rejection when  $\mu = 9.2$ ?

**Solution.** Since  $z_{.025} = 1.96$ ,  $z_{.25} = .67$ , the approximation from Equation (8.3.7) yields

$$n \approx \frac{(1.96 + .67)^2}{(1.2)^2} 4 = 19.21$$

Hence a sample of size 20 is needed. From Equation (8.3.4), we see that with  $n = 20$

$$\begin{aligned}\beta(9.2) &= \Phi\left(-\frac{1.2\sqrt{20}}{2} + 1.96\right) - \Phi\left(-\frac{1.2\sqrt{20}}{2} - 1.96\right) \\ &= \Phi(-.723) - \Phi(-4.643) \approx 1 - \Phi(.723) \\ &\approx .235\end{aligned}$$

Therefore, if the message is sent 20 times, then there is a 76.5 percent chance that the null hypothesis  $\mu = 8$  will be rejected when the true mean is 9.2. ■

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## 8.3.2 Case of unknown variance: the $t$ -test

$$H_0 : \mu = \mu_0$$

versus the alternative

$$H_1 : \mu \neq \mu_0$$

It should be noted that the null hypothesis is not a simple hypothesis since it does not specify the value of  $\sigma^2$ .



Now when  $\sigma^2$  is no longer known, it seems reasonable to estimate it by

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

and then to reject  $H_0$  when

$$\left| \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \right|$$

is large.

To determine how large a value of the statistic

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right|$$

to require for rejection, in order that the resulting test have significance level  $\alpha$ , we must determine the probability distribution of this statistic when  $H_0$  is true. However, as shown in Section 6.5, the statistic  $T$ , defined by

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S}$$

has, when  $\mu = \mu_0$ , a  $t$ -distribution with  $n - 1$  degrees of freedom. Hence,

$$P_{\mu_0} \left\{ -t_{\alpha/2, n-1} \leq \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \leq t_{\alpha/2, n-1} \right\} = 1 - \alpha \quad (8.3.11)$$

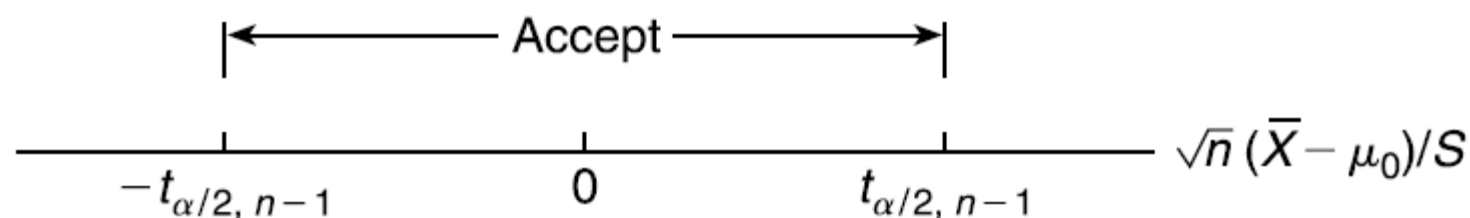
where  $t_{\alpha/2, n-1}$  is the 100  $\alpha/2$  upper percentile value of the  $t$ -distribution with  $n - 1$  degrees of freedom. (That is,  $P\{T_{n-1} \geq t_{\alpha/2, n-1}\} = P\{T_{n-1} \leq -t_{\alpha/2, n-1}\} = \alpha/2$  when  $T_{n-1}$  has a  $t$ -distribution with  $n - 1$  degrees of freedom.) From Equation (8.3.11) we see that the appropriate significance level  $\alpha$  test of

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

is, when  $\sigma^2$  is unknown, to

$$\text{accept } H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right| \leq t_{\alpha/2, n-1}$$

$$\text{reject } H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right| > t_{\alpha/2, n-1}$$



**Table 8.2**  $X_1, \dots, X_n$  Is a Sample from a  $(\mu, \sigma^2)$  Population,  $\sigma^2$  Is Unknown,  
 $\bar{X} = \sum_{i=1}^n X_i/n$   $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2/(n-1)$ .

| $H_0$                                                                                                          | $H_1$            | Test Statistic $TS$           | Significance Level $\alpha$ Test     | $p$ -Value if $TS = t$   |
|----------------------------------------------------------------------------------------------------------------|------------------|-------------------------------|--------------------------------------|--------------------------|
| $\mu = \mu_0$                                                                                                  | $\mu \neq \mu_0$ | $\sqrt{n}(\bar{X} - \mu_0)/S$ | Reject if $ TS  > t_{\alpha/2, n-1}$ | $2P\{T_{n-1} \geq  t \}$ |
| $\mu \leq \mu_0$                                                                                               | $\mu > \mu_0$    | $\sqrt{n}(\bar{X} - \mu_0)/S$ | Reject if $TS > t_{\alpha, n-1}$     | $P\{T_{n-1} \geq t\}$    |
| $\mu \geq \mu_0$                                                                                               | $\mu < \mu_0$    | $\sqrt{n}(\bar{X} - \mu_0)/S$ | Reject if $TS < -t_{\alpha, n-1}$    | $P\{T_{n-1} \leq t\}$    |
| $T_{n-1}$ is a $t$ -random variable with $n-1$ degrees of freedom: $P\{T_{n-1} > t_{\alpha, n-1}\} = \alpha$ . |                  |                               |                                      |                          |

# Testing the equality of means of two normal populations

**Table 8.4**  $X_1, \dots, X_n$  Is a Sample from a  $(\mu_1, \sigma_1^2)$  Population;  $Y_1, \dots, Y_m$  Is a Sample from a  $(\mu_2, \sigma_2^2)$  Population,  
The Two Population Samples Are Independent to Test  
 $H_0 : \mu_1 = \mu_2$  versus  $H_0 : \mu_1 \neq \mu_2$ .

| Assumption                 | Test Statistic $TS$                                                                       | Significance Level $\alpha$ Test       | $p$ -Value if $TS = t$     |
|----------------------------|-------------------------------------------------------------------------------------------|----------------------------------------|----------------------------|
| $\sigma_1, \sigma_2$ known | $\frac{\bar{X} - \bar{Y}}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}}$                            | Reject if $ TS  > z_{\alpha/2}$        | $2P\{Z \geq  t \}$         |
| $\sigma_1 = \sigma_2$      | $\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}} \sqrt{1/n + 1/m}}$ | Reject if $ TS  > t_{\alpha/2, n+m-2}$ | $2P\{T_{n+m-2} \geq  t \}$ |
| $n, m$ large               | $\frac{\bar{X} - \bar{Y}}{\sqrt{S_1^2/n + S_2^2/m}}$                                      | Reject if $ TS  > z_{\alpha/2}$        | $2P\{Z \geq  t \}$         |

# The paired t-test

$n$  pairs  $(X_i, Y_i), i = 1, \dots, n$  gas consumption of the  $i$ th car

$X_i$   $Y_i$

cannot treat  $X_1, \dots, X_n$  and  $Y_1, \dots, Y_n$  as independent samples

let  $\tilde{W}_i = X_i - Y_i, i = 1, \dots, n$ .

$H_0 : \mu_w = 0$  versus  $H_1 : \mu_w \neq 0$

where  $W_1, \dots, W_n$  are assumed to be a sample from a normal population having unknown mean  $\mu_w$  and unknown variance  $\sigma_w^2$ . But the  $t$ -test described in

accepting  $H_0$  if  $-t_{\alpha/2, n-1} < \sqrt{n} \frac{\bar{W}}{S_w} < t_{\alpha/2, n-1}$   
rejecting  $H_0$  otherwise

# Hypothesis tests concerning the variance

Let  $X_1, \dots, X_n$  denote a sample from a normal population  
unknown mean  $\mu$  and unknown variance  $\sigma^2$

test the hypothesis  $\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$

$$H_0 : \sigma^2 = \sigma_0^2$$

Because  $P\{\chi_{n-1}^2 < \chi_{\alpha/2, n-1}^2\} = 1 - \alpha/2$  and  $P\{\chi_{n-1}^2 < \chi_{1-\alpha/2, n-1}^2\} = \alpha/2$ , it follows that

versus the alternative

$$H_1 : \sigma^2 \neq \sigma_0^2$$

$$P_{H_0} \left\{ \chi_{1-\alpha/2, n-1}^2 \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi_{\alpha/2, n-1}^2 \right\} = 1 - \alpha$$

Therefore, a significance level  $\alpha$  test is to

$$\begin{array}{ll} \text{accept } H_0 & \text{if } \chi_{1-\alpha/2, n-1}^2 \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi_{\alpha/2, n-1}^2 \\ \text{reject } H_0 & \text{otherwise} \end{array}$$

$$p\text{-value} = 2 \min(P\{\chi_{n-1}^2 < c\}, 1 - P\{\chi_{n-1}^2 < c\})$$